

CHAPTER 5: **COMPUTER AND CYBER** **CRIME**

DR. ROZI NOR HAIZAN
BINTI NOR

OBJECTIVES

- Describe and categorize **cybercrimes or cyber-related crimes**
- Distinguish **cyber-related crimes and cyber-exacerbated crimes**
- Knows who the **perpetrators** are
- State the different **kinds of crimes**
- Know the related **Law**
- Know effort to **combat such crimes**

INTRODUCTION

■ Reported Incidents based on General Incident Classification Statistics 2020

#	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	TOTAL
Intrusion	122	93	125	144	0	0	0	0	0	0	0	0	484
Denial of Service	0	1	3	7	0	0	0	0	0	0	0	0	11
Content Related	23	23	42	23	0	0	0	0	0	0	0	0	111
Cyber Harassment	37	27	58	65	0	0	0	0	0	0	0	0	187
Intrusion Attempt	13	8	8	11	0	0	0	0	0	0	0	0	40
Vulnerabilities Report	5	7	10	10	0	0	0	0	0	0	0	0	32
Fraud	807	725	798	1,180	0	0	0	0	0	0	0	0	3,510
Malicious Codes	56	32	33	40	0	0	0	0	0	0	0	0	161
Spam	11	27	14	8	0	0	0	0	0	0	0	0	60
	1,074	943	1,091	1,488	0	0	0	0	0	0	0	0	4,596

MALAYSIA CYBERCRIME CASES

YEAR	TOTAL CASES	LOSSES (RM MILLION)
2007	1139	11.4
2008	1821	12.9
2009	3863	22.3
2010	6167	63.0
2011	6586	80.5
2012	4738	96.1

TOP 3 NUMBER OF CASES:

1. *E-COMMERCE FRAUD- ONLINE PURCHASE*
2. *PARCEL SCAM*
3. *VOIP SCAM- CROSS BORDER SYNDICATES*

DEFINING CYBERCRIME

- Confusion in labeling “computer crimes”
- computer crime ⇔ cybercrime
- Computer crime definition:
 - The act of criminal using **computer all a tool**



DEFINING CYBERCRIME

- (genuine) Cybercrime definition:

- A crime in which the criminal act can be carried out only through the use of **cybertechnology** and can take place only in the **cyberrealm**

CATEGORIES OF GENUINE CYBERCRIME

- Cyberpiracy : using **cybertechnology** in **unauthorized ways** to:
 - Reproduce copies or proprietary software and proprietary information or
 - Distribute proprietary information (in digital form) across a computer network
 - Eg: the distribution of MP3 files that contain proprietary material

CATEGORIES OF GENUINE CYBERCRIME

- Cybertrespass : using **cybertechnology** to gain **unauthorized access** to:
 - An individual's or an organization's computer system or;
 - A password-protected web site
 - Eg. Denial-of-service attacks on a e-commerce web site



CATEGORIES OF GENUINE CYBERCRIME

- Cyber vandalism : using **cybertechnology** to **unleash one or more programs** that:
 - Disrupt the transmission of electronic information across one or more computer networks, including the Internet, or
 - Destroy data resident in a computer or damage a computer system's resources, or both
 - Eg. Unleashing the ILOVEYOU virus

CYBER-EXACERBATED CRIMES

- Cyber-exacerbated crimes

- Cybertechnology does much more than merely assist someone in carrying out a crime
- Example : cyberstalking, internet pedophilia, internet pornography

CYBER-ASSISTED CRIMES

- Cyber-assisted crimes

- Cybertechnology merely assist someone in carrying out a crime
- Example : income tax cheating, physical attack with a computer, property damage using a computer

TYPE OF PERPETRATORS

■ Hackers

- Individuals who test the limitations of systems out of intellectual curiosity to see whether can gain access and how far they can go

■ Crackers

- Break into other people's networks and system, deface web pages, crash computers, spread harmful programs

TYPE OF PERPETRATORS

■ Malicious Insiders

- An ever-present adversary (number one concern)
- Estimated 70% network intruders come from inside:
- Employees, consultants, contractors
- Motivated by financial gain, revenge on employers, or publicity

TYPE OF PERPETRATORS

■ Malicious Insiders -- cont

- Has knowledge of system and authorized to access the systems
- Know procedure to gain access to login ids and password
- Know how systems work and where the weak points are
- Know about organizational structure and security procedure to avoid any investigation

TYPE OF PERPETRATORS

■ Industrial spies

- Use illegal means to obtain trade secrets from the competitors of the firm for which they are hire
- Theft of trade secret by disgruntled employees or ex-employees

TYPE OF PERPETRATORS

■ Cybercriminal

- Hack into corporate computer and steal money often transferring money from one account into another
- Engage in all forms of computer fraud
- Can spend large sum of money to buy expertise and access

TYPE OF PERPETRATORS

■ Cyberterrorists

- Someone who intimidates or coerces a government or organization to advance his or her political or social objectives by launching computer-based attacks against computers, networks, and information stored on them
- Cause harm rather than to gather information
- Result in outright destruction or disruption of services
- Can be well-organized, well-trained, and well-supported

FRAUD, EMBEZZLEMENT, SABOTAGE, IDENTITY THEFT, AND FORGERY

■ Some causes of **Fraud**

➤ **Credit-Card**

- ✓ Stolen receipts, mailed notices, and card
- ✓ Interception of online transaction or weak e-commerce security
- ✓ Careless handling by card-owner

➤ **ATM**

- ✓ Stolen account numbers and PINs
- ✓ Insider knowledge
- ✓ A counterfeit ATM

FRAUD, EMBEZZLEMENT, SABOTAGE, IDENTITY THEFT, AND FORGERY

■ Embezzlement and Sabotage

➤ Some causes

- ✓ Insider information
- ✓ Poor security
- ✓ Complex financial transaction
- ✓ Anonymity of computer users

➤ Some defenses

- ✓ Rotate employee responsibility
- ✓ Require use of employee ID and password
- ✓ Implement audit trail
- ✓ Careful screening and background checks of employees



FRAUD, EMBEZZLEMENT, SABOTAGE, IDENTITY THEFT, AND FORGERY

■ Identity Theft

➤ **Some causes**

- ✓ Insecure and inappropriate use of identification card number
- ✓ Careless handling of personally identifiable information
- ✓ Weak security of stored records
- ✓ Insufficient assistance to identity theft

➤ **Some defenses**

- ✓ Limit use of personally identifiable information
- ✓ Increase security of information stored by business and government agencies
- ✓ Improve methods to accurately identify a person
- ✓ Educate consumers

FRAUD, EMBEZZLEMENT, SABOTAGE, IDENTITY THEFT, AND FORGERY

■ Forgery

➤ **Some causes**

- ✓ Powerful computers and digital manipulation software
- ✓ High-quality printers, copiers, and scanners

➤ **Some defenses**

- ✓ Educate consumers and employees
- ✓ Use anti-counterfeiting techniques during production
- ✓ Use counterfeit detection methods
- ✓ Create legal and procedural incentives to improve security

CYBERCRIME IN MALAYSIA

- Cybercrime has **surpassed** drug trafficking as the most lucrative crime.
- Almost **anybody** who is an active computer/ online user would have been a cybercrime victim, and in most cases too its perpetrators.
- The **younger generation** are the most vulnerable.
- Use of technology to cheat, harass, disseminate false information.
- **70 %** of commercial crime cases now can be categorised as **cybercrime** cases.

Ref:
Cybercrime: Malaysia Document
DSP Mahfuz bin Dato' Ab. Majid
Royal Malaysia Police

CURRENT CHALLENGES IN CYBERCRIME INVESTIGATION : ROYAL MALAYSIAN POLICE

- Mobility of devices proving to be biggest challenge to law enforcement in tracking and analysing equipment.
- Introduction of new technologies and economic innovations. Ensuring adequate analytical and technical capabilities for law enforcement .
- Maintaining a comprehensive and relevant legal framework.
- Working in a borderless environment with laws of multiple jurisdictions. Issue of differences in national priorities.
- Implementing information security practices and raising awareness.

COMPUTER CRIMES ACT 1997

- Date of coming into operation: 1/6/2000
- **Purpose:** to provide punishment related to computer misuse
- **The law gives enforcer a framework to define:**
 - Reading, overtaking, and using computer together with computer information illegally
 - Standard for service providers
 - Outlines punishment to those guilty
- **The mentioned crimes include:**
 - Trespassing
 - Steal and destroy data
 - Deface of web pages
 - Steal e-mail
 - Misuse of credit card for fraud

COMPUTER CRIMES ACT 1997

- **Cybercrime on property :**

- steal information, money, service.

- Consists of fraud, forgery, mischief

- **Examples are:**

- Using statement and wrongly interpret to persuade the country to give property and valuables such as multilevel marketing, franchise, auction through computer

- Spreading of computer virus through downloaded programs or email attachment

- Illegal access computer systems with intention to destroy or disrupt information

COMPUTER CRIMES ACT 1997

- **Cybercrime on human:**

- **Examples are:**

- Pornography, cyber disturbance in which victims are disturbed through internet via web pages, e-mail or chat
- Stalking to threaten victims via e-mail message
- Illegal access computer systems with intention to destroy or disrupt information

COMPUTER CRIMES ACT 1997

- **Cyber terrorist**

- **Examples are:**

- Accessing air traffic control to detour flight route to cause crashes
- Remote access to food manufacturing system by changing the level of iron that can cause death to children

COMPUTER CRIMES ACT 1997

■ **Three mains' offences:**

- **Unauthorized access to computer material, Section 3**
- **Unauthorized access with intent to commit or facilitate commission of further offence, Section 4**
- **Unauthorized modification of the contents of any computer, Section 5**

COMPUTER CRIMES ACT 1997

■ **Unauthorized access to computer material:**

- Section 3(1), a person shall be guilty of an offence if:
 - ✓ He causes a computer to perform any function with intent to secure access to any program or data held in any computer
 - ✓ The access he intends to secure is unauthorized; and
 - ✓ He knows at the time when he causes the computer to perform the function that is the case.
- Fine not exceeding RM50,000 or imprisonment for a term not exceeding five years or both

COMPUTER CRIMES ACT 1997

- **Unauthorized access with intent to commit or facilitate commission of further offence**

- Section 4, a person shall be guilty of an offence if he commits an offence referred to in section 3 with intent:
 - ✓ To commit an offence involving fraud or dishonesty or which causes injury as defined in the Penal Code [Act 574]; or
 - ✓ To facilitate the commission of such an offence whether by himself or by any other person
 - ✓ It is immaterial whether the offence is to be committed at the same time when the unauthorized access is secured or on any future occasion
- Fine not exceeding RM150,000 or imprisonment for a term not exceeding ten years or both

COMPUTER CRIMES ACT 1997

■ **Unauthorized modification of the contents of any computer:**

- Section 5, a person shall be guilty of an offence if he does any act which he knows will cause unauthorized modification of the contents of any computer
- ✓ Section 2(7), a modification of the contents of any computer takes place if, by the operation of any function of the computer concerned or any other computer -
 - a) any program or data held in the computer concerned is altered or erased
 - b) any program or data is introduced or added to its contents; or
 - c) any event occurs which impairs the normal operation of any computer
- ✓ And any act that contributes towards causing such modification shall be regarded as causing it

COMPUTER CRIMES ACT 1997

- **Unauthorized modification of the contents of any computer**
- Any modification is authorized if, 2(8)
 - a) a person whose act causes it is not himself entitled to determine whether the modification should be made; and
 - b) he does not have consent to the modification from any person who is so entitled
- ✓ And any act that contributes towards causing such modification shall be regarded as causing it

COMPUTER CRIMES ACT 1997

■ **Wrongful communication, Section 6**

- A person shall be guilty of an offence if he communicates directly or indirectly a number, code, password or other means of access to a computer to any person other than a person to whom he is duly authorized to communicate
- Fine not exceeding RM25,000 or imprisonment for a term not exceeding three years or both

COMPUTER CRIMES ACT 1997

■ **Abetments and attempts punishable as offences, Section 7**

- A person who abets the commission of or who attempts to commit any offence shall be guilty of that offence
- A person who does any act preparatory to or in furtherance of the commission of any offence shall be guilty of that offence

COMPUTER CRIMES ACT 1997

■ **Presumption, Section 7**

- A person who has in his custody or control any program, data or other information which is held in any computer or retrieved from any computer which he is not authorized to have in his custody or control shall be deemed to have obtained unauthorized access to such program, data or information unless the contrary is proved

CYBERLAWS OF MALAYSIA

- 1.COMPUTER CRIME ACT 1997
- 2.COMMUNICATIONS AND MULTIMEDIA ACT 1998 (CMA)
- 3.MALAYSIAN COMMUNICATIONS AND MULTIMEDIA COMMISSION ACT 1998
- 4.DIGITAL SIGNATURE ACT 1997
- 5.COPYRIGHT ACT (AMENDMENT) 1997
- 6.TELEMEDICINE ACT 1997
- 7.OPTICAL DISC ACT 2000
- 8.ELECTRONIC TRANSACTIONS ACT 2006

Souce:ROYAL MALAYSIA POLICE

**EXAMPLE CYBERCRIME
CASE IN MALAYSIA
OFFENCES UNDER
COMPUTER CRIME ACT
1997**

CASE-1

PP v Nebolisa Olisa Hillary [2016]

Nebolisa, a Nigerian was accused of cheating a retiree via Facebook, after the latter befriended him who posed as an Arab-English male. When he informed her about his “financial problem”, she transferred RM10,7950 to a Maybank account in multiple transactions on 7 December 2015 and between 8 to 15 December 2015.

Charges:

Section 420 Penal Code (2 charges) (pleaded guilty).

Decision of Batu Pahat Magistrate Court (11/02/2016):

18 months imprisonment and 2 strokes of cane for each charge.

CASE-2

PP v Lau Jia Wen [2018]

Lau, 23, an electronics and furniture company employee was accused of modifying the advertised price of iPhone 7 Plus from RM4299 to RM4.99 on the website of the company without authority with the intention to cheat on 22 July 2017.

Charges:

Section 5(1) Computer Crimes Act.

Decision of Kuala Lumpur Sessions Court (16/05/2018):

RM3000 fine.

CASE-3

Multiple Offences under the Penal Code and Computer Crimes Act 1997

PP v Rose Hanida bt Long [2016]
Rose Hanida bt Long v PP [2017] MLJU 1212

Rose Hanida, 36, a secretary to the Head of Corporate Banking Department at OCBC Bank was accused of 13 series of unauthorised access using ID and password belonging to the Department Head. The acts were committed to cheat the OCBC Bank Finance Department by submitting false financial claims, causing the department to approve and deposit RM348,294.81 to her account in 160 transactions, between January 2010 to December 2013.

CASE-3 CONT

Charges:

Section 4(1)(a) Computer Crimes Act (13 charges).

Section 420 Penal Code (13 charges).

Decision of Kuala Lumpur Sessions Court (30/08/2016):

1 year imprisonment for each charge under section 4(1)(a).

RM15,000 fine for each charge under section 4(1)(a).

4 years imprisonment for each charge under section 420.

RM 5,000 fine for each charge under section 420.

She appealed against the sentences.

OFFENCES UNDER THE COMMUNICATION AND MULTIMEDIA ACT 1998

CASE-1

PP v Syahzan Amir Endut [2019]

Syahzan, 41, an employee of Universiti Utara Malaysia was accused of disseminating an obscene photo of his ex-girlfriend via MMS on 28 March 2015.

Charge:

Section 233(1)(a) CMA.

Decision of Kulim Sessions Court (26/09/2018):

RM15000 fine.

The prosecution appealed against the sentence.

Decision of Alor Setar High Court(22/07/2019):

Dismissed the appeal by the prosecution.

CASE-2

PP v Aszroy bin Achoi [2018] 9 MLJ 702
PP v Aszroy bin Achoi [2019]

Aszroy, 25, was accused of supporting Daesh via his two Facebook accounts “Yohyo Irranun Peace AlBorneozy” and “Yohyo Illa’nun AlSaba Malizia” between 8 May to 19 July 2016. He was also accused of having possession in his mobile phone 43 images associated with Daesh.

Charges:

Section 130J(1)(a) Penal Code.

Section 130JB(1)(a) Penal Code.

CASE-2...CONT

Decision of Kota Kinabalu High Court (28/10/2017):

7 years imprisonment for the first charge.

2 years imprisonment for the second charge.

The sentences were ordered to run consecutively from the date of arrest.

The prosecution appealed against sentence.

Decision of the Court of Appeal (18/03/2019):

Dismissed the appeal.

CASE-3

PP v Chow Mun Fai [2019]

Chow Mun Fai, a hawker was accused of inciting racial tension between Malays and non-Malays by insulting the death of fire fighter Muhammad Adib Mohd Kassim on Twitter using the profile name of “Alvin@Alvin15358333” on 13 February 2019. He was also accused of uploading an image insulting Prophet Muhammad, which may incite disharmony between Muslims and non-Muslims on 2 March 2019.

He was also accused of posting other offensive tweets using the same profile.

Muhammad Adib died on 17 December 2018 after being admitted to the National Heart Institute due to severe injuries sustained during a riot near Maha Mariamman Temple in Seafield on 26 November 2018.

CASE-3 ..CONT

Charges:

Section 505 Penal Code.

Section 298A(1)(a) Penal Code (3 charges).

Section 233(2) CMA (4 charges).

(pleaded guilty to all charges).

Decision of Kuala Lumpur Sessions Court (06/08/2019):

7 to 30 months imprisonment for all the charges.

ALL OF CYBERCRIME CASES ARE TAKEN FROM
[HTTPS://WWW.AMAZON.COM/CYBERCRIME-CASES-
DECADEMALAYSIAN-EXPERIENCE/DP/1692537709/](https://www.amazon.com/CYBERCRIME-CASES-DECADEMALAYSIAN-EXPERIENCE/DP/1692537709/)

Biography of author:



Rizal Rahman is an Associate Professor at the Faculty of Law, Universiti Kebangsaan Malaysia. His areas of expertise are Cyber Law, Evidence Law, Constitutional and Administrative Law, and Business Law.

MALAYSIA IS SIXTH MOST VULNERABLE TO CYBER CRIME

10 Riskiest Countries

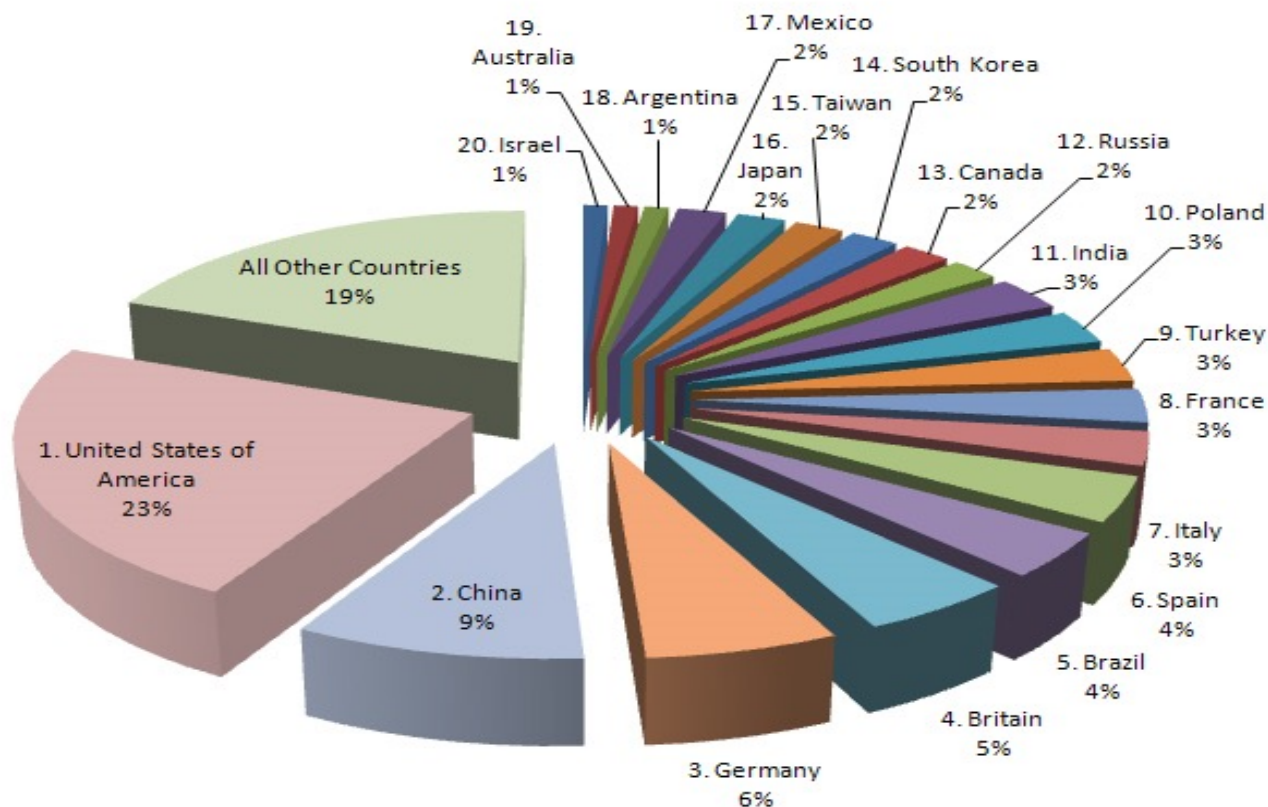
	TER		TER
1. Indonesia	23.54%	6. India	15.88%
2. China	21.26%	7. Mexico	15.66%
3. Thailand	20.78%	8. UAE	13.67%
4. Philippines	19.81%	9. Taiwan	12.66%
5. Malaysia	17.44%	10. Hong Kong	11.47%

Threat exposure rate (TER): Measured as the percentage of PCs that experienced a malware attack, whether successful or failed, over a three month period.

Source: SophosLabs

says.com

TOP 20 COUNTRIES WITH THE HIGHEST RATE OF CYBERCRIME



Cybercrime: Top 20 Countries

(source: BusinessWeek/Symantec)